

DENTINE TUTORIAL

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TASK 1: IDENTIFY, PASTE, STUDY AND LABEL THE FOLLOWING FIGURES

Note: please write the figure heading and labellings by
hand in your logbooks.

Fig. 9.3: Ground section showing dentinal tubules cut longitudinally showing sinusoidal primary curvatures.



Fig 9.4: Ground, partially demineralised section of dentinal tubules cut transversely.

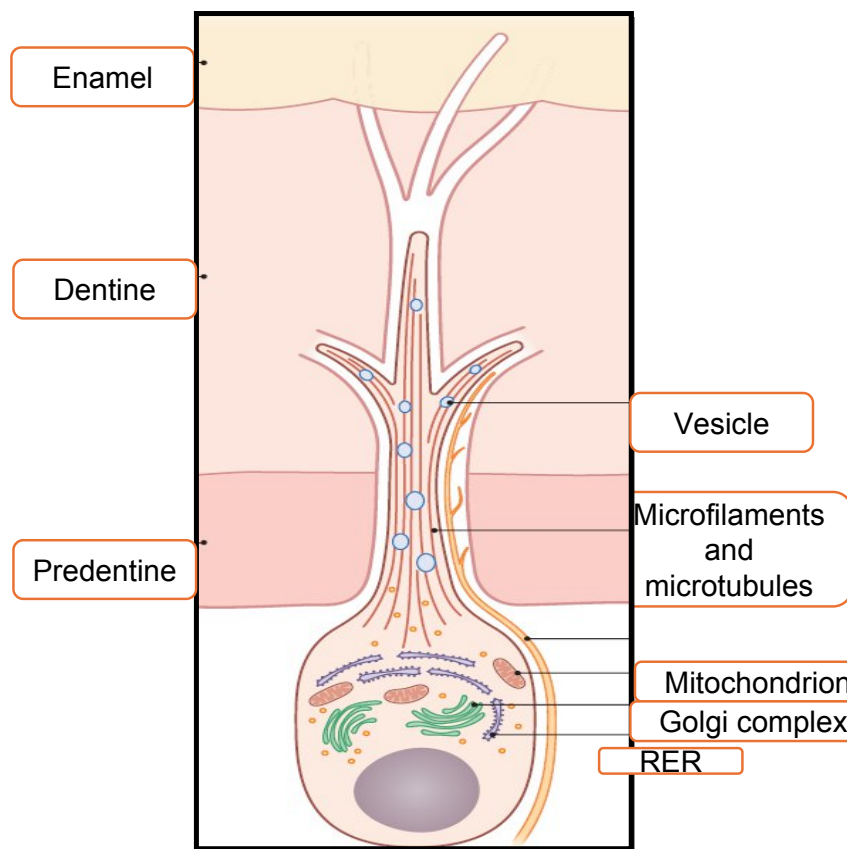
A-Perimeter of a tubule

B-Intertubular dentine

C-Peritubular dentine



Fig 9.19: Diagram of an odontoblast and its main process extending into the dentine.



**Fig 9.31: Regions of
varying dentine structure.**

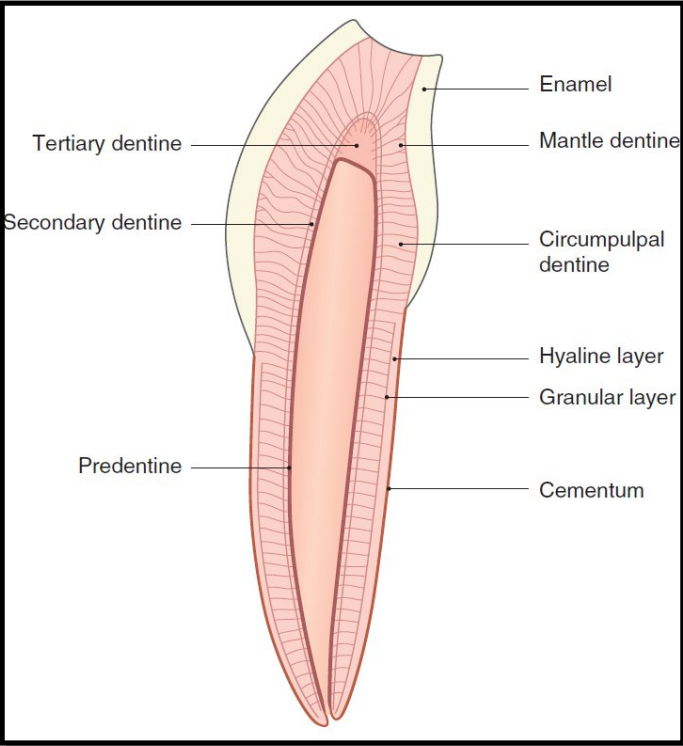
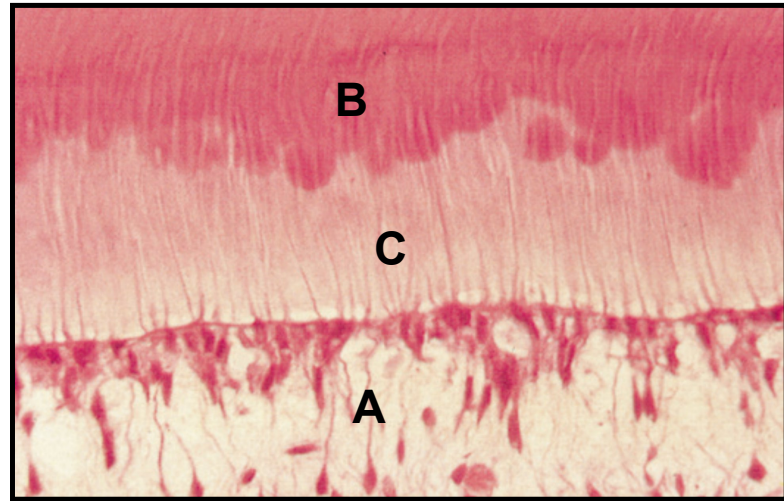


Fig 9.39: Decalcified section of the pulpodentinal region showing odontoblast layer.

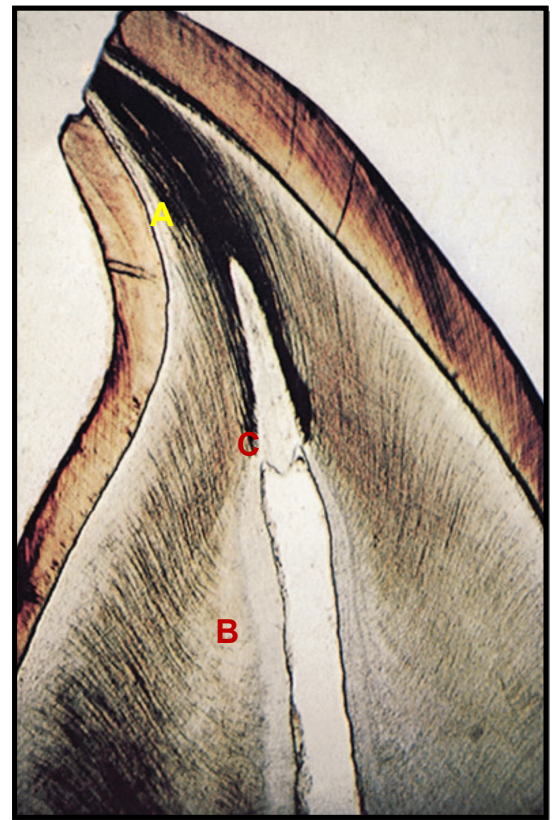
A- Odontoblast layer

B- Mineralised dentine matrix

C- Predentine

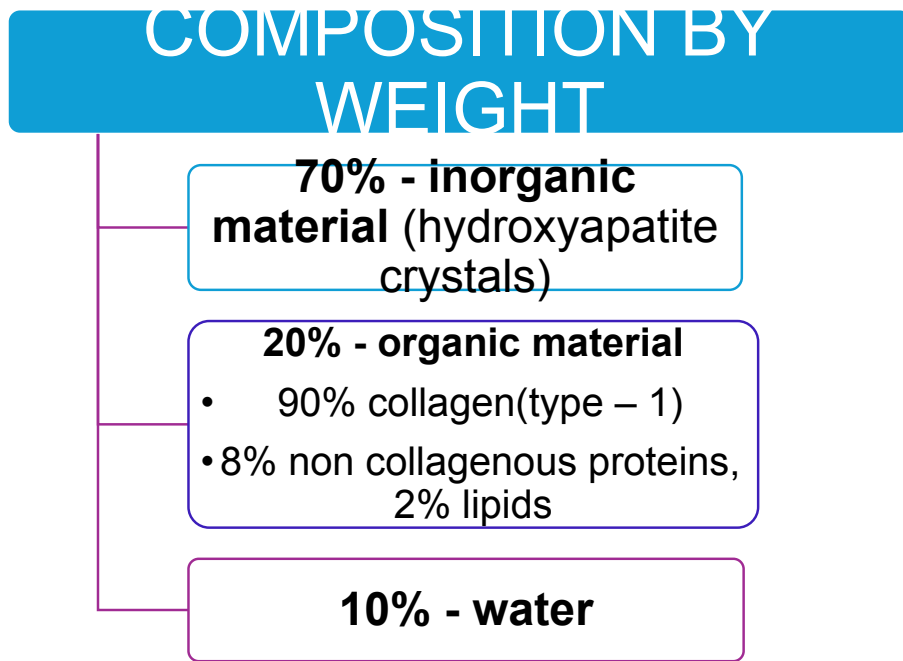


9.62: Ground longitudinal section of a crown showing a dead tract (A) beneath a region of attrition that is sealed pulpally by tertiary dentine (C). Secondary dentine (B) lines the rest of the pulp.



TASK 2: ANSWERS THE QUESTIONS

Q1) State the composition of dentin?



Q2) Differentiate between mantle and circumpulpal dentin?

Mantle Dentin	Circumpulpal Dentin
Forms outer layer of dentin	Forms bulk of the inner dentin towards pulp.
5% less mineralized	It is more mineralized
Collagen fibres are oriented $\perp$ to DEJ	Collagen fibres are oriented $\parallel$ to DEJ
Dentinal tubules profusely branch in mantle dentin	No such profuse branching is evident in circumpulpal dentin
Mineralization with matrix vesicles	Mineralization by heterogenous nucleation

Q3) What are dentinal tubules? Describe it's features?

Ans. Dentine is permeated by the dentine tubules that run from the pulpal surface to the DEJ and CDJ.

FEATURES:

- 1- The dentin tubules follow a curved, sigmoid course, primary curvatures.
- 2- In the root and beneath the cusps, the primary curvatures are less pronounced, the tubules running a straighter course.
- 3- The tubules taper from approximately 2.5 μm in diameter at their pulpal ends to 1 μm or less peripherally.
- 4- The tubules also show changes in direction of much smaller (a few micrometers) amplitude. These are known as the secondary curvatures.
- 5- Dentinal tubules branch. The most profuse branching is in the periphery near the DEJ.

Q5) What are the structural lines in dentine and features.

Ans. There are two related groups of lines: 1. originating from curvatures in the dentinal tubules. 2. Arising from incremental deposition of dentine and its subsequent mineralization.

LINES ASSOCIATED WITH THE PRIMARY CURVATURES OF THE DENTINAL TUBULES

- They would be best seen in longitudinal section as broad concentric bands in the circumpulpal dentine. These lines are known as [Schreger lines](#).

LINES ASSOCIATED WITH THE SECONDARY CURVATURES OF THE DENTINAL TUBULES

- When the secondary curvatures coincide they also give rise to an optical effect, resulting in the appearance of lines, the [contour lines of Owen](#).
- The [neonatal line](#) may be also included in it.

INCREMENTAL LINES ASSOCIATED WITH MATRIX DEPOSITION AND MINERALIZATION

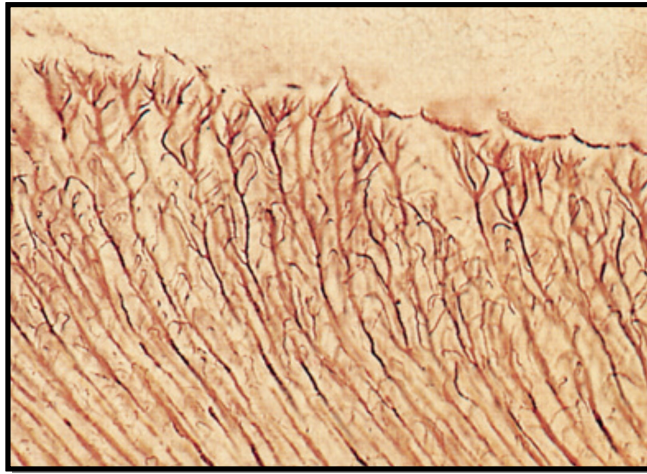
- Dentine has regular, incremental, short-period and long-period markings.
- Short-period markings may be seen as alternating dark and light bands, each pair reflecting the diurnal rhythm of dentine formation. These fine lines are sometimes referred to as [von Ebner's lines](#).
- In cuspal dentine- distance between adjacent dark bands is approximately 4 μm .
- In the root dentine- distance between lines is nearer 2 μm .
- The coarser, long-period lines ([Andresen lines](#)) are approximately 16–20 μm apart . Between each long-period line there are six to 10 pairs of short-period lines.

STUDY MATERIAL FOR REFERENCE ONLY

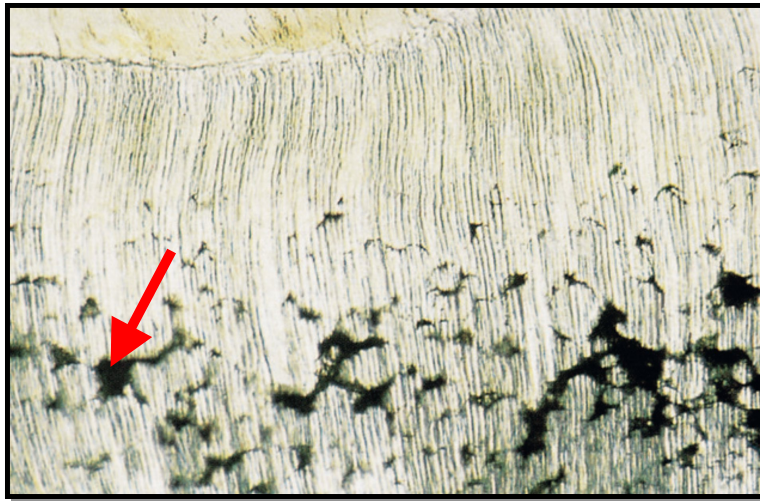
Note: This is for study purposes only, no need to write in logbooks.

C: Peritubular dentine is highly mineralized dentine, formed along the walls of dentinal tubules. It lacks collagenous fibrous matrix. It narrows the lumen and its formation gradually leads to obliteration of the tubule.

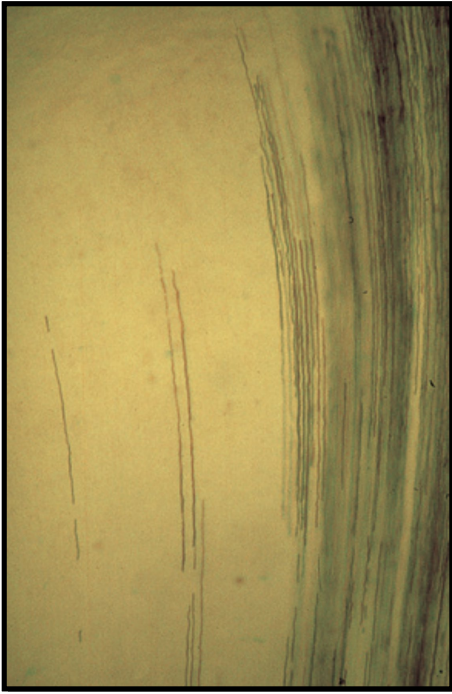




Ground section of dentine near the enamel–dentine junction (arrow) showing profuse branching of dentinal tubules (mantle dentine).



Interglobular dentin describes areas of unmineralized or hypomineralized dentine. When globules (zones of mineralization) do not fuse properly into a homogenous mass within mature dentin, it results in the formation of interglobular dentine.

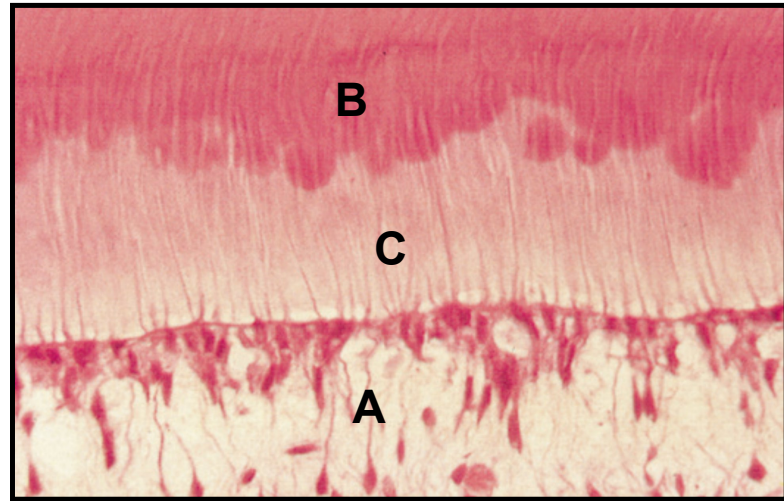


Sclerotic dentine is a type of dentine in which the dentinal tubules become filled with mineral deposits, making them less permeable and more translucent. It appears glass-like or transparent in ground sections

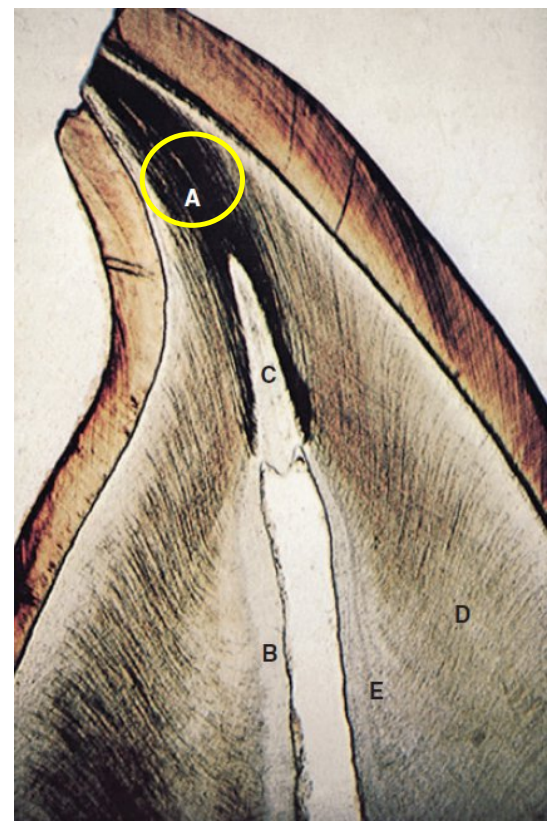
B: The mineralizing front may show a globular or a linear outline, reflecting the mineralization.

C: Predentine appears pale, being unmineralized, width varying 10 μm to 40 μm . It is rich in type I collagen.

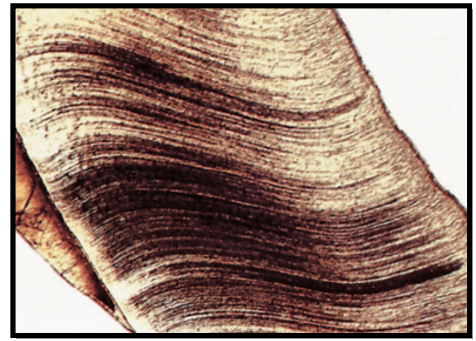
A: Role of the odontoblast in predentine is secretion of matrix components. In mineralising dentine, it is to participate in the modification of that matrix.



A: Dead tracts are the areas in dentine where the odontoblastic processes have degenerated or died because of insult/trauma, or retract before peritubular dentine occludes the tubules, leaving the dentinal tubules empty. These empty tubules contain air in the ground sections and therefore appear dark.



Primary curvatures: Due to the movement of odontoblasts after the formation of dentine, the tubules do not run straight. They follow S-shaped curvature



Secondary curvatures: Small, wave-like bends resulting from the lateral movement of odontoblasts. These are superimposed on the primary curvature.

